Investigation of Tungsten Thermophotovoltaic Cells in Geothermal Settings

Background:

As of 2025, 74% of U.S. energy is generated from fossil fuels, making it the second biggest carbon contributor globally (ClientEarth Communications, 2025). Additionally, new U.S. policies are loosening environmental limitations that were once in place for companies in regards to energy sourcing, meaning the energy market is starting to lend towards fossil fuels once again. To compete with the convenience of nonrenewable energy, green energy systems constantly need to find new ways to be more efficient and cheaper. Using thermophotovoltaic (TPV) cells, an emerging energy solution, geothermal energy production can double its efficiencies and become a steady contender for renewable energy generation. TPV systems are being optimized in various lab settings and have shown promising potential for implementation in actuality. This type of system requires further research, but relies on the geological heat powering the thermal emitter, which radiates photons to a photovoltaic cell to generate electricity. Choosing proper materials, site location, system setup, fabrication schematics, and data manipulation, a blueprint can be created for a geothermal system to contribute to the implementation of this design in the future.

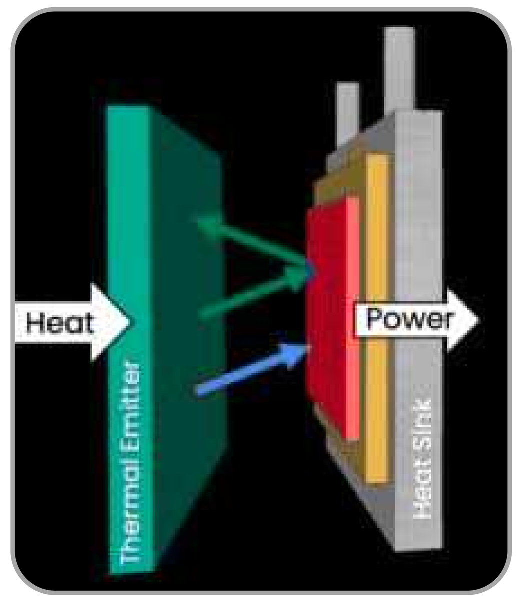

Figure 1- TPV System Diagram


Figure 2 - TPV System

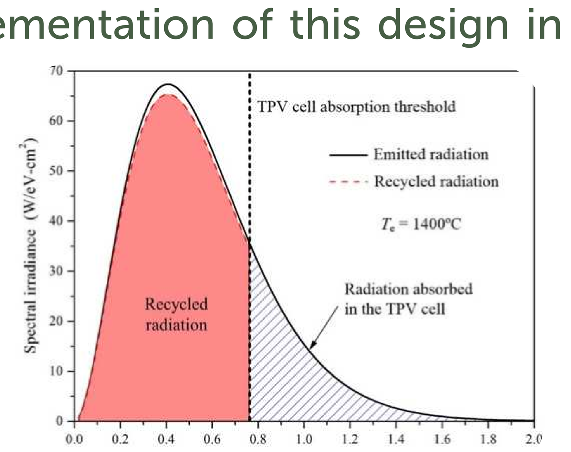

Figure 3 - Photon Recycling Graph

Deliverables:

Theoretical analysis was necessary to complete this investigation, due to the complex fabrication that comes with creating a TPV cell. Cells can not be purchased because they are not commercially available at this point, and only reside in collegiate studies currently. For that reason, physical prototypes were not possible within the scope of this project.

Something that can be studied and contributed to the field of thermophotovoltaics, is a white paper, or comparative analysis of the system in a new light or looking at it from a new perspective. Comparisons are made between the proposed solution and some current renewable energy sources. The comparison is on behalf of efficiency, long-term durability, and other determining factors in regard to clean energy sourcing.

Methods:

PHASE ONE

TPV Emitters and emitter materials
- suitability for geothermal settings

TPV semiconductors and semiconductor materials
- manipulation of bandgap, past sustainability

Spectral Filtering
- nanofilters and spectral recyclers based on efficiency increase

Thermal Radiation Modeling for Geothermal Environments
- radiation characteristics at different depths, heat optimized geothermal sites

Thermal Management/Cooling
- current management techniques and efficiencies


Figure 4 - TPV System MIT

PHASE TWO

Bandgap width selection (suiting semiconductor and emitter)
- bandgap ranges, manipulation difficulty of each

Bandgap manipulation for selected emitter and semiconductor materials

Finding a spectral filter to recycle photons not in bandgap
- temperature constraints

Estimated temperature and geological level for the heat pump

Geographical location for thermal site

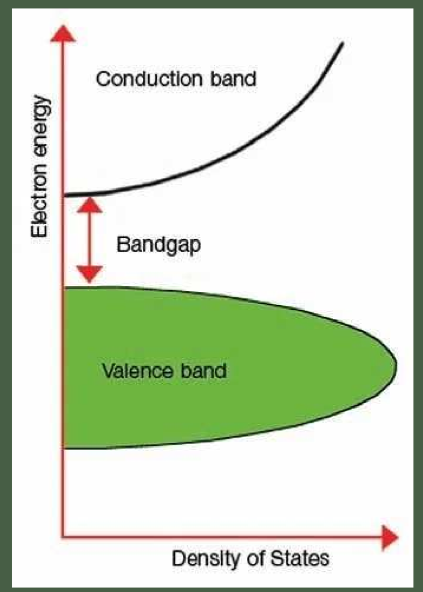

Figure 5 - Bandgap Efficencey Chart

PHASE THREE

Define 2 baseline systems which the solution can be compared to
- establishes a reference point for data and comparison for data outputs

Assess how aspects of the experiment are changing and how that would affect experimental values

Aspects being altered and what that does in regards to the TPV system
- report how conditional changes effect proposed solution vs. actual systems

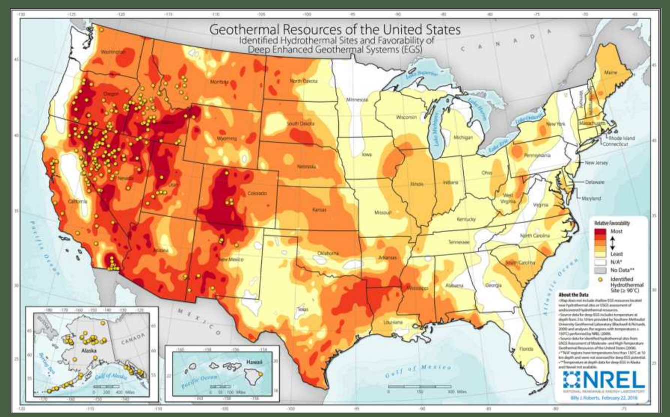

Figure 6 - Heat Map of Geothermal Potential

PHASE FOUR

Research on other geothermal energy extraction methods/heat recyclers
- to be used as a benchmark for the TPV system outputs

Find quantitative data on the 3 chosen competitors
- measurable data is used to ground the TPV system predictions

Longevity and sustainability of the systems role in the final analysis
- relates the data back to immeasurable categories

How does cost and manufacturing play a role in the feasibility of these systems
- nonquantifiable category which is still very relevant to the analysis

Conclusion:

The material properties of tungsten, particularly its high melting point and thermal stability, make it well-suited for the extreme conditions of geothermal environments. When paired with carefully selected semiconductors and optimized band gap alignment, TV systems can significantly improve energy capture from infrared radiation that would otherwise be lost.

Findings from this study suggest that spectral filtering, emitter optimization, and effective thermal management are critical to maximizing system performance. While current limitations in fabrication, cost, and material compatibility remain barriers to immediate large-scale implementation, the theoretical models and comparative analyses indicate strong potential for future development.

Overall, tungsten TV systems could serve as a complementary technology within geothermal energy infrastructure, increasing total energy output and contributing to more efficient and sustainable power generation. Continued research and technological advancements will be essential in bridging the gap between conceptual design and practical application.

Further Research:

If resources were available, next steps could be to create a software to model the TV system and produce blueprint data points for quantitative comparison to current systems, which would promote the fabrication and implementation of this system commercially.

Alternatively, if other materials were selected for further research, comparison could occur between the solution cell with Tungsten and Gassing, and another cell made of other materials.

The one component of this project that would require the most further research, however, is the spectral filtering. Fabrication for this component is not only tideous, but requires specific adjustment based on the molecular structure and composition of the material piece used, which was not achievable within the theoretical design of this solution.

Citations:

Experimental performance comparison of 0.72 EV-gasb and 0.59 EV-ingaas thermophotovoltaic cells under different radiation temperatures - sciencedirect. (n.d.-a). https://www.sciencedirect.com/science/article/abs/pii/S0306261924003428

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